	Case Name: TEMPORARY WORKS **Project Owner:** Eder Omar Vilca Carrillo	Sector	**Construction**
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Processed by	**Zahra Hussaini**		User defined distributions
Date	**2025**	Project Control	Automatic tracking
File Name	**C2025-17**		Tracking based on user input

1. Project description

Project authenticity

This project describes the different activities and their interrelations in the improvement and rehabilitation of infrastructure, sanitary systems, electrical installations, and environmental protection works at and around a thermal bath facility, including support for health and hygiene services, safety measures, and COVID-19 prevention, aiming to enhance user experience, accessibility, and safety conditions in the area. The project consists of activity and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	170
Planned Duration (PD)	90 days
Budget At Completion (BAC)	338686.53 €
Renewable Resources	8 Types
Consumable Resources	0 Types

Network topology	
Serial/Parallel (SP)	26%
Activity Distribution (AD)	47%
Length of Arcs (LA)	0%
Topological Float (TF)	29%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0.6	7.7	13.0
CRI-rho	51.5	3.7	13.0
CRI-tau	99.8	2.0	-13.0

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	7.7	11.7	1.2
SI	16.7	15.1	1.2
SSI	1.1	7.4	9.8
CRI-r	8.2	7.4	3.4
CRI-rho	8.1	7.4	3.1
CRI-tau	9.2	7.3	1.8

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	0	0	N/A
CRI-tau	0	0	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section "2.2. Risk Analysis". Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	13.7	13.0
PV - SPI	14.5	13.3
PV - SCI	14.5	13.3
ED - 1	11.9	11.4
ED - SPI	13.0	11.7
ED - SCI	13.0	11.7
ES - 1	12.0	11.5
ES - SPI(t)	13.3	12.1
ES - SCI(t)	13.3	12.1

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0	0
CPI	0	0
SPI	1.5	0.3
SPI(t)	2.1	0.5
SCI	1.5	0.3
SCI(t)	2.1	0.5
0.8 CPI + 0.2 SPI	0.3	0.1
0.8 CPI + 0.2 SPI(t)	0.5	0.2

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 3 tracking periods. The Real Duration and Real Cost mentioned in section "2.3.3. Earned Value Management" are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-1389.5	-107399.8	-285.7	1.0	0.5	0.3	0.7
std dev	2463.0	127706.3	341.6	0.0	0.4	0.5	0.5
final	-4233.2	0.0	-81.0	1.0	1.0	0.9	1.0

2.3.3.2. Time forecasting

PD	90 days

Real Duration	102 days

Late	12%

EAC(t)			**Real Accuracy**	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	120.3	34.9	6.0	-6.0
PV - SPI	298.0	188.0	64.1	-64.1
PV - SCI	298.7	186.9	64.3	-64.3
ED - 1	123.0	37.2	6.9	-6.9
ED - SPI	306.7	183.9	66.9	-66.9
ED - SCI	306.7	183.9	66.9	-66.9
ES - 1	127.0	42.4	8.2	-8.2
ES - SPI(t)	335.7	403.9	76.4	-76.4
ES - SCI(t)	335.7	403.9	76.4	-76.4

2.3.3.3. Cost forecasting

BAC	338686.53 €

Real Cost	338686.53 €

On budget	0%

EAC			**Real Accuracy**	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	340140.8	2463.0	0.4	-0.4
CPI	340081.2	2517.3	0.4	-0.4
SPI	993481.9	635630.7	193.3	-193.3
SPI(t)	980497.0	1107872.1	189.5	-189.5
SCI	993258.0	635614.5	193.3	-193.3
SCI(t)	979977.0	1106971.5	189.3	-189.3
0.8 CPI + 0.2 SPI	374639.9	28919.9	10.6	-10.6
0.8 CPI + 0.2 SPI(t)	357903.1	29581.6	5.7	-5.7